Biomedicine & Pharmacotherapy 145 (2022) 112410

Contents lists available at ScienceDirect

Biomedicine & Pharmacotherapy

journal homepage: www.elsevier.com/locate/biopha

Anti-asthmatic effects of *Phlomis umbrosa* Turczaninow using ovalbumin induced asthma murine model and network pharmacology analysis

So-Won Pak [a,1], A Yeong Lee [b,1], Yun-Soo Seo [b], Se-Jin Lee [a], Woong-Il Kim [a], Dong-Ho Shin [a], Jong-Choon Kim [a], Joong-Sun Kim [a], Je-Oh Lim [a,*], In-Sik Shin [a,*]

[a] *College of Veterinary Medicine and BK21 FOUR Program, Chonnam National University, 77 Yongbong-ro, Buk-gu, Gwangju 61186, South Korea*
[b] *Herbal Medicine Resources Research Center, Korea Institute of Oriental Medicine, 111 Geonjae-ro, Naju-si, Jeollanam-do 58245, South Korea*

A R T I C L E I N F O

Keywords:
Phlomis umbrosa Turczaninow
Asthma
Network analysis
ERK
MMP-9

A B S T R A C T

Background: *Phlomis umbrosa* Turczaninow has been used as a tradition herbal medicine for treating various inflammatory diseases.
Purpose: In present study, we explored the effects of *P. umbrosa* on asthma induced by ovalbumin (OVA) and elucidated the mechanism via in vivo verification and network pharmacology prediction.
Methods: The animals were intraperitoneally injected OVA on day 1 and 14, followed by OVA inhalation on days 21, 22, and 23. The animals were daily treated *P. umbrosa* extract (PUE, 20 and 40 mg/kg) by oral gavage from day 18 to day 23.
Results: PUE significantly decreased airway hyperresponsiveness, eosinophilia, and the production of inflammatory cytokines and OVA specific immunoglobulin E in animals with asthma, along with a reduction in airway inflammation and mucus secretion in lung tissue. In network analysis, antiasthmatic effects of PUE were closely related with suppression of mitogen-activated protein kinases and matrix metalloproteinases (MMPs). Consistent with the results from network analysis, PUE suppressed the phosphorylation of ERK and p65, which was accompanied by a decline in MMP-9 expression.
Conclusion: Administration of PUE effectively reduced allergic responses in asthmatic mice, which was associated with the suppressed phosphorylation of ERK and p65, and expression of MMP-9. These results indicate that PUE has therapeutic potential to treat allergic asthma.

1. Introduction

Asthma is triggered by environmental factors such as airborne allergens (e.g., animal dander, dust mites, pollen), irritants in the air as a result of air pollution (e.g., chemical fumes, smoke), certain foods and food additives, extreme weather conditions, and even stress. The global prevalence has been increasing every year [11]. Asthma is a chronic airway inflammatory disease with specific clinical manifestations such as shortness of breath, chest pain, and cough [3]. The immunopathogenesis of asthma involves inflammatory cytokines such interleukin (IL)-4, -5, and -13 that act as initiators, this is associated with the eosinophilic inflammation, production of immunoglobulin (Ig)E and chemokines, mucus production, and airway hyperresponsiveness (AHR) [21,28]. Based on pervious literatures, drugs that inhibit the production of inflammatory cytokines have been developed for asthma treatment. However, these drugs are limited in their use owing to adverse effects or low efficacy [19]. Accordingly, there is an increasing demand for the development of drugs with no adverse effects and high efficacy for proper management of asthma.

Phlomis umbrosa Turczaninow is a traditional herbal medicine

Abbreviations: ADME, absorption, distribution, metabolism and excretion; AHR, airway hyperresponsiveness; BALF, bronchoalveolar lavage fluid; DAVID, database for annotation, visualization, and integrated discovery; DL, drug-likeness; ELISA, enzyme-linked immunosorbent assay; ERK, extracellular signal-regulated kinase; HPLC, High-performance liquid chromatography; KEGG, Kyoto encyclopedia of genes and genomes; Ig, immunoglobulin; IL, interleukin; MMP-9, matrix metalloproteinases-9; NF-κB, nuclear factor-kappa B; OB, oral bioavailability; OVA, ovalbumin; PBS, phosphate-buffered saline; PPI, protein-protein interaction; PUE, *Phlomis umbrosa* Turczaninow ethanol extract; TCMSP, traditional Chinese medicine systems pharmacology database.

* Corresponding authors.
E-mail addresses: 166634@jnu.ac.kr (J.-O. Lim), dvmshin@jnu.ac.kr (I.-S. Shin).
[1] These authors have equally contributed in this work.

https://doi.org/10.1016/j.biopha.2021.112410
Received 30 September 2021; Received in revised form 2 November 2021; Accepted 4 November 2021
Available online 11 November 2021

described in the 'Dong-Eui-Bo-Gam' written by Jun Heo in 1613 and is used to treat inflammation, pain, and phlegm. A few experimental studies have also demonstrated pharmacological properties of *P. umbrosa* that help to increase bone density and promote bone growth [13,18,26]. Additionally, it is reported that *P. umbrosa* has an anti-allergic effect on mast cell-dependent allergic reaction [23,24]. Based on these studies, it is considered that *P. umbrosa* effectively protects against allergic airway inflammation of asthma. However, there have been no studies examining whether *P. umbrosa* are effective for asthma.

Therefore, we investigated the efficacy of *P. umbrosa* in the treatment of asthma using a murine model of ovalbumin (OVA)-induced asthma. We performed high performance liquid chromatography (HPLC) to analyze the active ingredients of *P. umbrosa*, and western blot and network pharmacology analysis to understand its mechanism of action.

2. Materials and methods

2.1. Preparation of P. umbrosa and analysis of its active ingredients

2.1.1. Plant

P. umbrosa was obtained from MyRyeung Herbal Medicine (Pocheon-si, South Korea). Plant extracts were obtained after confirmation of morphological and genetic variations by Dr. Goya Choi and Dr. Byeong Cheol Moon at Korea Institute of Oriental Medicine (KIOM), respectively. A voucher specimen (No. 2-17-0072) was deposited in the Korean Herbarium of Standard Resources, KIOM (Daejeon, South Korea). This sample was refluxed in 70% ethanol (v/v) for 2 h and the yield of 70% ethanol extract of *P. umbrosa* (PUE) was 26.62% (w/w).

2.2. Chemicals

Sesamoside, shanzhiside methyl ester, and umbroside (= 8-acetyl shanzhiside methyl ester) were obtained from ChemNorm Biotech Co., Ltd (Wuhan, Hubei, China), Biopurify Phytochemicals Ltd. (Chengdu, Sichuan, China), and ChemFaces Biochemical Co., Ltd. (Wuhan, Hubei, China), respectively.

2.3. HPLC analysis

PUE (111.12 mg) was dissolved in 10 mL of 70% ethanol and filtered using a syringe filter before for HPLC analysis. HPLC system (Waters, Milford, MA) consisted of Acquity QDa detector (Waters), Micro-splitter (IDEX Health & Science LLC, Oak Habor, WA), Separation Module (Waters e2695), and 2998 PDA detector (Waters). Three components in 70% ethanol extracts were analyzed using the XSelect™ HSS T3 column (5 μm, 4.6 × 250 mm, Waters). Mobile phase was prepared by mixing 0.05% aqueous formic acid (A) and acetonitrile (B), and a linear gradient program was set up from 95% A to 75% A for 30 min. Flow rate was 0.8 mL/min, injected volume was 10 μL, and column temperature was ambient. UV wavelength was monitored from 200 to 400 nm, and three target peaks were detected at 237 nm. QDa condition was set up as the follows: nitrogen as the carrier gas, positive/negative TIC mode, ESI capillary at 0.80 kV, probe temperature at 600 °C, Con Voltage of 15 V, source temperature at 121 °C, and 20:1 split.

2.4. Animal experiment

2.4.1. Experimental procedure

BALB/c female mice (6 weeks old, 19–20 g, SAMTAKO, Osan, South Korea) were housed under normal conditions as follows: 22 ± 2 °C, 55 ± 5% relative humidity, and a 12 h night/day cycle. They were provided feed and water ad libitum. The protocol of animal experiments was approved by the Chonnam National University Institutional Animal Care and Use Committee (CNU IACUC-YB-2016–19, Gwangju, South Korea). To induce asthma, on days 1 and 14, mice were administered OVA (20 μg, Sigma-Aldrich, St. Louis, MO) with aluminum hydroxide (30 mg, Sigma-Aldrich) intraperitoneally. Following this, the animals were inhaled with 1% OVA for 1 h from day 21–23 using ultrasonic nebulizer (Omron, Tokyo, Japan). PUE (20 and 40 mg/kg) and montelukast (30 mg/kg) were orally administered to the mice from day 18–23. We measured the AHR of each animal on day 24, and then the animals were sacrificed on day 25. Five experimental groups (n = 5) were designed as follows: NC (normal animals with oral administration of PBS), OVA (asthma model with oral administration of PBS), MON (asthma model with oral administration of montelukast), and PUE 20 and 40 (asthma model with oral administration of PUE (20 and 40 mg/kg, respectively)). On day 24, AHR was determined using whole-body plethysmography (All medicus, Seoul, South Korea). Each mouse was placed in a chamber and measured bronchoconstriction expressed as the dimensionless parameter—enhanced pause (Penh) following inhalation of PBS and increasing concentrations of methylcholine (Sigma-Aldrich) in PBS for 3 min. The highest value during each methylcholine dose exposure was expressed. On day 25, bronchoalveolar lavage fluid (BALF) was collected as described previously [4]. An endotracheal tube was inserted into the trachea of animal after tracheostomy. PBS (0.7 mL) was instilled into the lungs and then retracted. The collected samples were transferred to tubes. This process was repeated once more (total volume was 1.4 mL). The BALF samples were centrifuged (200 *g*, 10 min); the supernatant was stored at − 20 ℃ for later measuring cytokines, and the cell pellet was used for the measurement of inflammatory cell count. To separate out the serum, blood obtained from the caudal vena cava was centrifuged (200 *g*, 20 min).

2.5. Analysis of BALF and serum

The total cell count in BALF was measured by a cell countess II (Thermo Fisher Scientific, Waltham, MA). The differential inflammatory cell count in BALF was performed as our previous study [19]. BALF supernatant was to measure the production of IL-4, -5, and -13 using ELISA kits (R&D System, Minneapolis, MN). The quantification of OVA-specific IgE in serum was performed using ELISA in accordance with a previous study [19].

2.6. Histopathology

The left lung tissue was fixed in neutralized formalin, embedded, and sectioned to obtain 4 μm tissue sections. Tissue sections were stained with hematoxylin and eosin (Sigma-Aldrich) and periodic acid-Schiff (IMEB Inc., San Marcos, CA) to evaluate inflammatory response and mucus secretion, respectively. Quantification of inflammation and mucus secretion was determined using an image analyzer (IMT i-Solution Inc., Vancouver, BC, Canada). To determine the expression of MMP-9 in lung tissue, immunohistochemistry was performed as described previously [5]. The primary antibody was Anti-mouse MMP-9 antibody (1:200 dilution, Santa Cruz Biotechnology, Dallas, TX) was used as the primary antibody.

2.7. Western blot

To measure change in protein expression caused by administration of PUE, right lung tissue was homogenized (1/10 w/v) using a homogenizer in Tissue lysis/extraction regent (Sigma-Aldrich) supplemented with protease inhibitor (Sigma-Aldrich). Western blotting was performed in accordance with previous studies [11,20]. The primary antibodies used were as follows: phosphorylated-ERK (Abcam, Cambridge, UK), ERK (Abcam), phosphorylated p65 (Cell Signaling, Beverly, MA), p65 (Cell Signaling), and β-actin (Cell Signaling). Densitometric value of protein band was determined by ChemiDoc (Bio-Rad Laboratories, Hercules, CA).

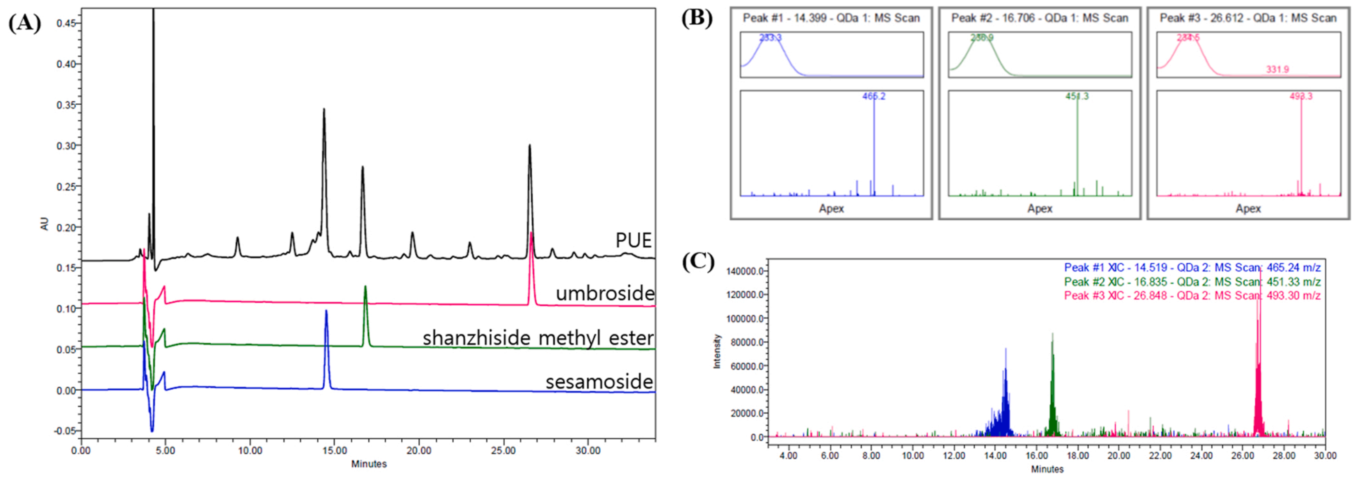

Fig. 1. HPLC chromatogram of PUE and standard components at 237 nm (a), verification of peak comparing λ_{max} and m/z of standard components; sesamoside (λ_{max} = 233.3, m/z = 465.2), shanzhiside methyl ester (λ_{max} = 236.9, m/z = 451.3), and umbroside (λ_{max} = 234.5, m/z = 493.3) (b), overlay of SIR peaks; sesamoside (blue, m/z = 465.2), shanzhiside methyl ester (green, m/z = 451.3), and umbroside (pink, m/z = 493.3) (c).

2.8. Statistical analysis

All data included in the analysis are represented as mean ± standard deviation. Statistical evaluation was performed using analysis of variance followed by Dunnett's post-hoc adjustments. P values < 0.05 and < 0.01 were considered as significant.

2.9. Network pharmacology analysis

2.9.1. Searching compounds in P. umbrosa

Compounds in *P. umbrosa* were searched PubMed (https://pubmed.ncbi.nlm.nih.gov/), Web of Science (https://www.webofknowledge.com), and TM-MC: A database of medicinal materials and chemical compounds in Northeast Asian traditional medicine (informatics.kiom.re.kr/compound/, updated in 8 April 2021). The compounds of information including 2D/3D structures, synonyms, chemical number, and physicochemical properties were identified to PubChem (https://pubchem.ncbi.nlm.nih.gov/, accessed in 28 July 2021) and ChemSpider (http://www.chemspider.com, version 2021.0.18.0) [14].

2.10. Active compounds through ADME screening

All compounds were filtered through an *in silico* integrative model Absorption, Distribution, Metabolism, and Excretion (ADME) using Traditional Chinese Medicine Systems Pharmacology Database (TCMSP; https://old.tcmsp-e.com/tcmsp.php, version 2.3) [8,12]. Among the various values provided by TCMSP, only oral bioavailability (OB) and drug-likeness (DL) were selected as active compounds. The values were OB ≥ 20% and DL≥ 0.10 according to TCMSP criteria [29]. OB is one of the most important pharmacokinetic parameters, and a high OB is often a key indicator of the DL value of a bioactive molecule [7,25]. DL is an indicator that is used to assess whether a compound has the potential to

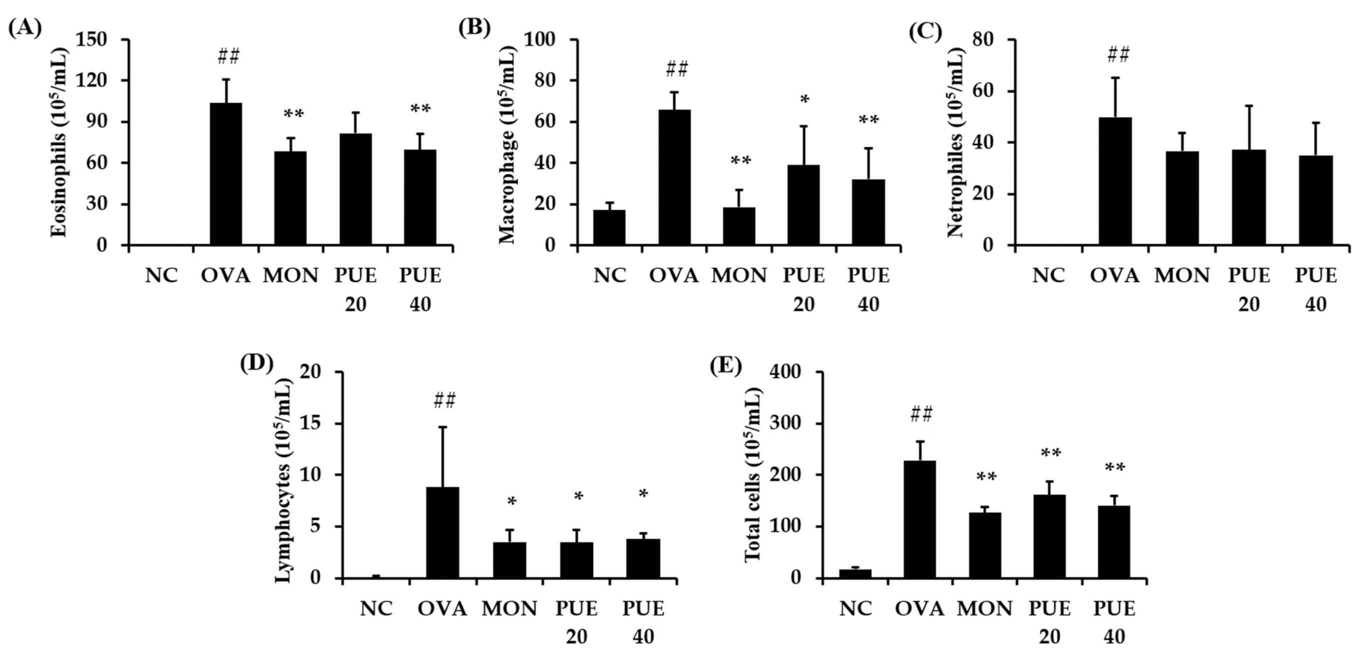

Fig. 2. PUE decreased inflammatory cell count of BALF in animals with asthma. (A) eosinophils, (B) macrophages, (C) neutrophils, (D) lymphocytes, (E) total cells. NC: PBS treatment and PBS challenge; OVA: PBS treatment and OVA challenge; MON: montelukast (30 mg/kg) treatment and OVA challenge; PUE 30 and 40: PUE (20 mg/kg and 40 mg/kg, respectively) and OVA challenge. The values are shown as the mean ± SD (n = 5). $^{\#\#}p < 0.01$ versus NC; *, $^{**}p < 0.05$ and 0.01 versus OVA.

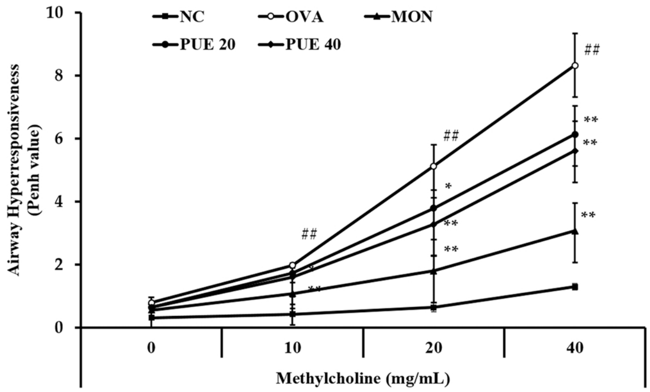

Fig. 3. PUE decreased AHR in animals with asthma. NC: PBS treatment and PBS challenge; OVA: PBS treatment and OVA challenge; MON: montelukast (30 mg/kg) treatment and OVA challenge; PUE 30 and 40: PUE (20 mg/kg and 40 mg/kg, respectively) and OVA challenge. The values are shown as the mean ± SD (n = 5). $^{\#\#}p < 0.01$ versus NC; *, $^{**}p < 0.05$ and 0.01 versus OVA.

be developed into a drug with respect to its physical and chemical properties; it is calculated based on the Tanimoto coefficient and relevant molecular descriptors [8,11].

2.11. Potential target genes

The target genes linked to active compounds were searched for the STITCH database (http://stitch.embl.de/, version 5.0). This DB provides a confidence score, which means that the higher value, the higher the correlation value between the compound and the gene [14]. Herein, confidence score was screened in a medium confidence score (≥ 0.400) with the '*Homo sapiens*' species setting. Human gene information of asthma was extracted from the GeneCards; The Human Gene Database (https://www.genecards.org/, version 5.4). Potential target gene was defined as genes that overlap the aforementioned target genes and the asthma-related genes in GeneCards DB.

2.12. Protein-Protein interaction (PPI)

The PPI network can predict protein complexes and functions of unknown proteins. Sophisticated network-based tools have been found to predict potential disease genes [10,25]. PPI analyses were conducted with the STITCH database and the medium confidence score (≥ 0.400), and the topology analysis was performed using Cytoscape version 3.7.2 (https://cytoscape.org/) [2,28].

2.13. Signal pathway analysis

Signal pathway analysis were conducted using the Database for Annotation, Visualization, and Integrated Discovery (DAVID) ver. 6.8 (https://david.ncifcrf.gov/, accessed in 9 August, 2021) and Kyoto Encyclopedia of Genes and Genomes (KEGG; https://www.genome.jp/kegg/, updated in 2, August, 2021). KEGG pathways with $p > 0.05$ were considered statistically significant [8,14]. And network was visualized using Cytoscape version 3.7.2.

3. Results

3.1. HPLC analysis of PUE

Fig. 1 confirmed the presence of the three components of *Phlomis umbrosa* Turczaninow—(1) umroside, (2) shanzhiside methyl ester, and (3) seamoside—at a wavelength of 273 nm using HPLC. Each component was measured at 27, 17, and 14.5 min

3.2. Animal experiment

3.2.1. PUE reduced inflammatory cell counts of BALF from OVA-induced asthmatic animals

Animals in the OVA group displayed noticeable elevation in inflammatory cell counts including eosinophils, macrophages, and

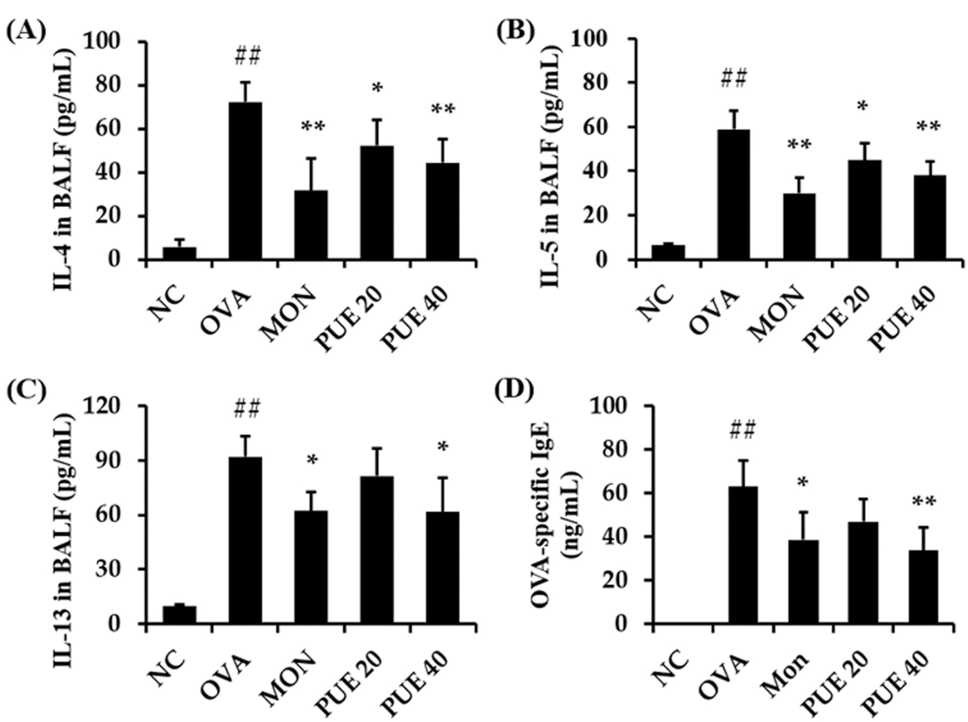

Fig. 4. PUE reduced the production of cytokines and OVA-specific IgE in animals with asthma. (A) IL-4, (B) IL-5, (C) IL-13, (D) OVA-specific IgE. NC: PBS treatment and PBS challenge; OVA: PBS treatment and OVA challenge; MON: montelukast (30 mg/kg) treatment and OVA challenge; PUE 30 and 40: PUE (20 mg/kg and 40 mg/kg, respectively) and OVA challenge. The values are shown as the mean ± SD (n = 5). $^{\#\#}p < 0.01$ versus NC; *, $^{**}p < 0.05$ and 0.01 versus OVA.

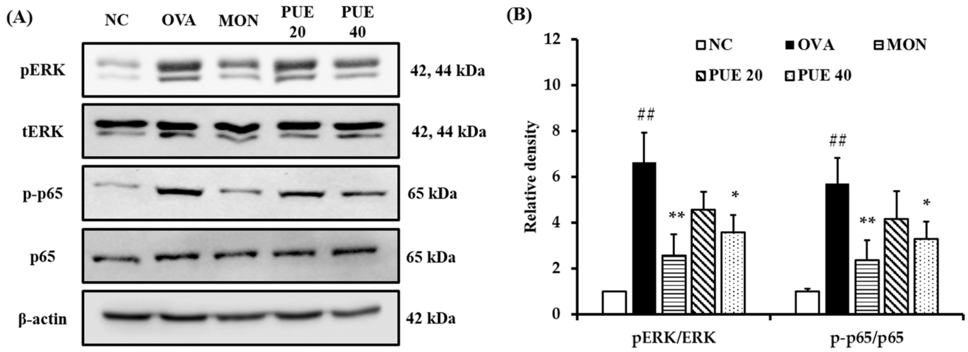

Fig. 5. PUE suppressed phosphorylation of ERK and p65 in animals with asthma. (A) the expression of ERK and p65, (B) relative density. NC: PBS treatment and PBS challenge; OVA: PBS treatment and OVA challenge; MON: montelukast (30 mg/kg) treatment and OVA challenge; PUE 30 and 40: PUE (20 mg/kg and 40 mg/kg, respectively) and OVA challenge. The values are shown as the mean ± SD (n = 5). $^{\#\#}p < 0.01$ versus NC; *, $^{**}p < 0.05$ and 0.01 versus OVA.

lymphocytes in comparison to those in the NC group (Fig. 2). However, montelukast treatment significantly decreased the number of eosinophils, macrophages, and lymphocytes in animals with asthma. Additionally, PUE treated groups (PUE 20 and 40) also showed marked decrease in the cell count of eosinophils, macrophages, and lymphocytes in OVA-induced asthmatic animals. In particular, the PUE 40 group exhibited significant decline in inflammatory cell counts compared with the OVA group.

3.2.2. PUE decreased AHR of OVA-induced asthmatic animals

AHR, an important index of asthma, was noticeably elevated in the OVA group more than the NC group (Fig. 3). By contrast, the MON group exhibited significant decrease in AHR in comparison to the OVA group. In addition, PUE treated groups (PUE 20 and 40) showed marked decrease in AHR compared with the OVA group. Particularly, the PUE 40 group displayed significant decline in AHR compared with the OVA group with increasing concentrations of methylcholine.

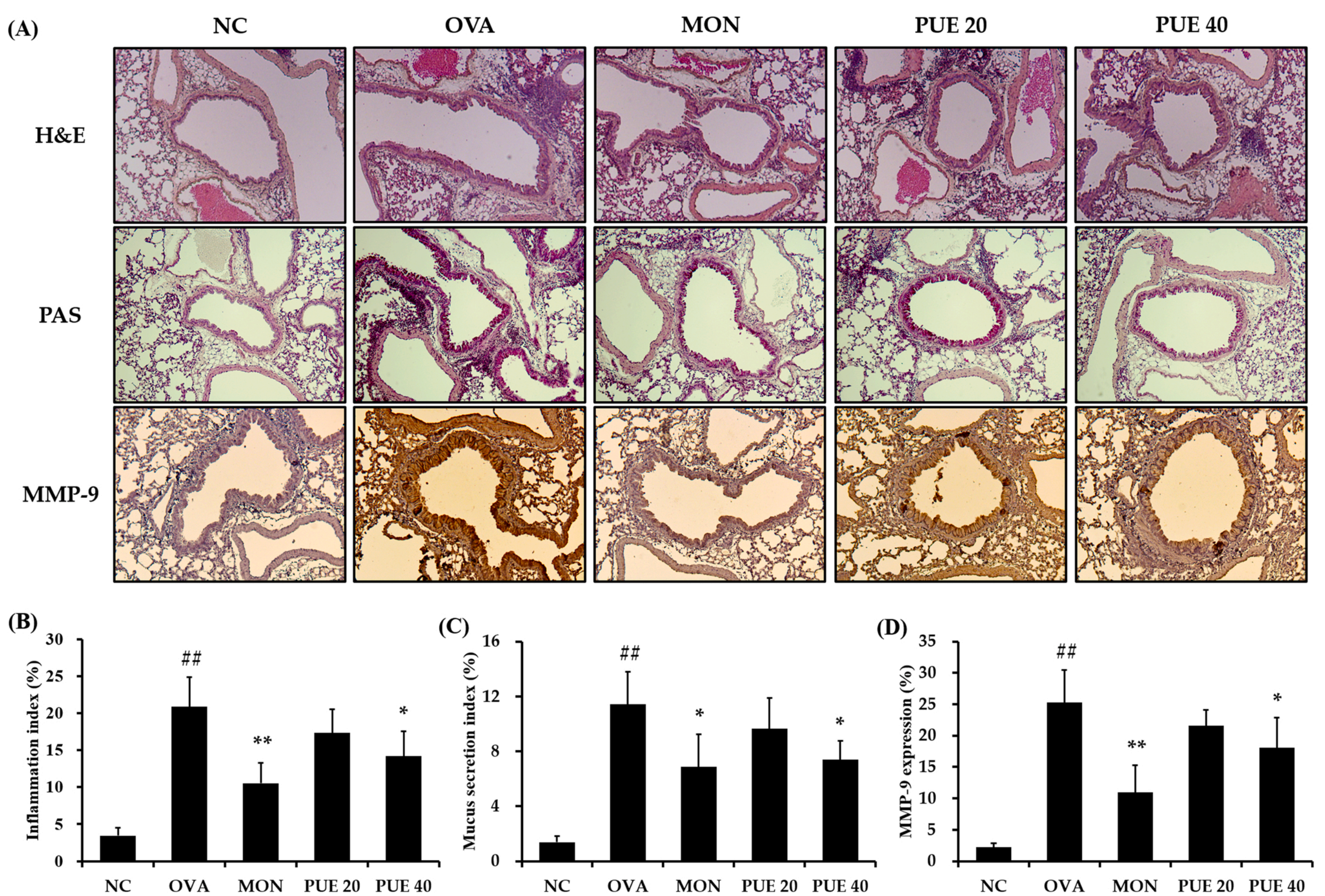

Fig. 6. PUE attenuates pulmonary inflammation, mucus secretion, and MMP-9 expression in animals with asthma. (A) representative figure of H&E, PAS and IHC (MMP-9), (B) inflammation index, (C) mucus secretion index, (d) MMP-9 expression value. NC: PBS treatment and PBS challenge; OVA: PBS treatment and OVA challenge; MON: montelukast (30 mg/kg) treatment and OVA challenge; PUE 30 and 40: PUE (20 mg/kg and 40 mg/kg, respectively) and OVA challenge. The values are shown as the mean ± SD (n = 5). $^{\#\#}p < 0.01$ versus NC; *, $^{**}p < 0.05$ and 0.01 versus OVA.

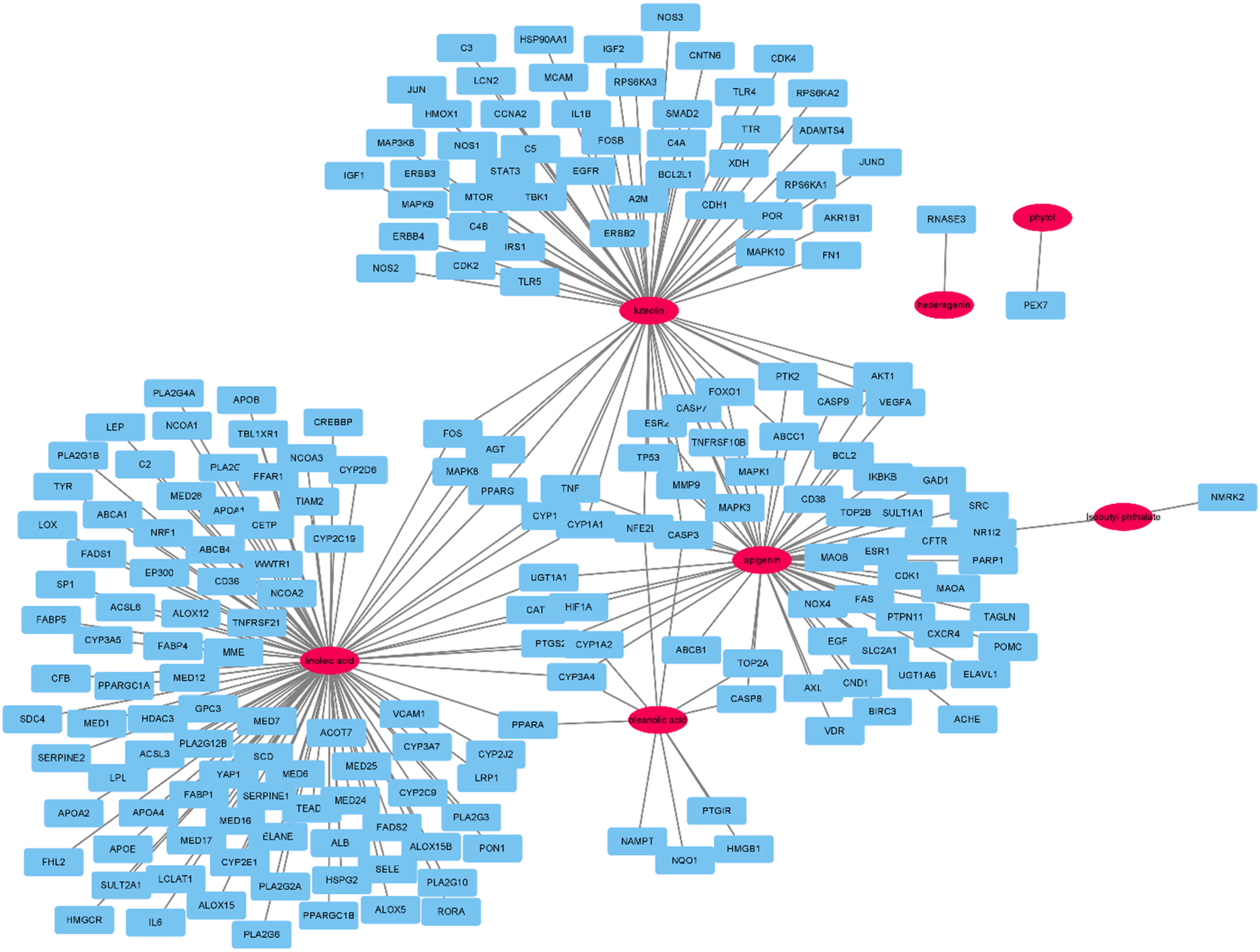

Fig. 7. Network of seven compounds (pink oval) and 202 potential target genes (cyan rectangles): this network was comprised of 209 nodes and 240 edges.

3.2.3. PUE reduced the production of cytokines and OVA-specific IgE in OVA-induced asthmatic animals

The OVA group displayed elevated levels of cytokines IL-4, -5, and -13 in BALF compared with the NC group (Fig. 4A–C, respectively). In contrast, mice in the MON group had decreased levels of these cytokines in BALF compared with those in the OVA group. In addition, PUE treated groups (PUE 20 and 40) also had reduced levels of cytokines in BALF compared with those in the OVA group; particularly, the PUE 40 group showed an even greater reduction in cytokine levels. Similar to cytokines, OVA-specific IgE in serum was also noticeably elevated in mice of the OVA group compared with those in the NC group, while PUE treated groups (PUE 20 and 40) showed significant decrease in OVA-specific IgE level compared with those in the OVA group (Fig. 4D).

3.2.4. PUE suppressed phosphorylation of ERK and p65 in OVA-induced asthmatic animals

The phosphorylation level of ERK was noticeably higher in OVA group animals than those in the NC group (Fig. 5). In contrast, animals in the MON group exhibited decreased ERK phosphorylation in comparison to those in the OVA group. PUE treated groups (PUE 20 and 40) also exhibited reduced ERK phosphorylation compared with those in the OVA group. Consistent with ERK phosphorylation, OVA group also exhibited markedly elevated levels of phosphorylation of p65 compared with those in the NC group, whereas it was significantly declined in the PUE treated groups in comparison to the OVA group.

3.2.5. PUE attenuated pulmonary inflammation, mucus secretion, and MMP-9 expression in OVA-induced asthmatic animals

OVA group animals exhibited prominent increase in pulmonary inflammation and mucus secretion in comparison to those in the NC group (Fig. 6). In contrast, MON group had markedly reduced pulmonary inflammation and mucus secretion. In addition, PUE treated groups (PUE 20 and 40) also exhibited suppressed pulmonary inflammation and mucus secretion, and in particular, PUE 40 showed an even higher reduction than the OVA group. Similarly, MMP-9 expression was higher in the OVA group than the NC group, whereas it was significantly declined in the PUE treated groups (PUE 20 and 40) in comparison to those in the OVA group.

3.3. Network pharmacology analysis

3.3.1. Active compounds

Total 84 compounds were selected through searching literatures and public DB. List of compounds were shown in Supplementary Table S1. Finally, 14 compounds were selected as active compounds through ADME screening. *In silico* modeling of ADME properties has been proven to be useful in the search for active compounds, which are mainly responsible for the therapeutic effects of a given drug [9]. List of active compounds were represented in Supplementary Table S2.

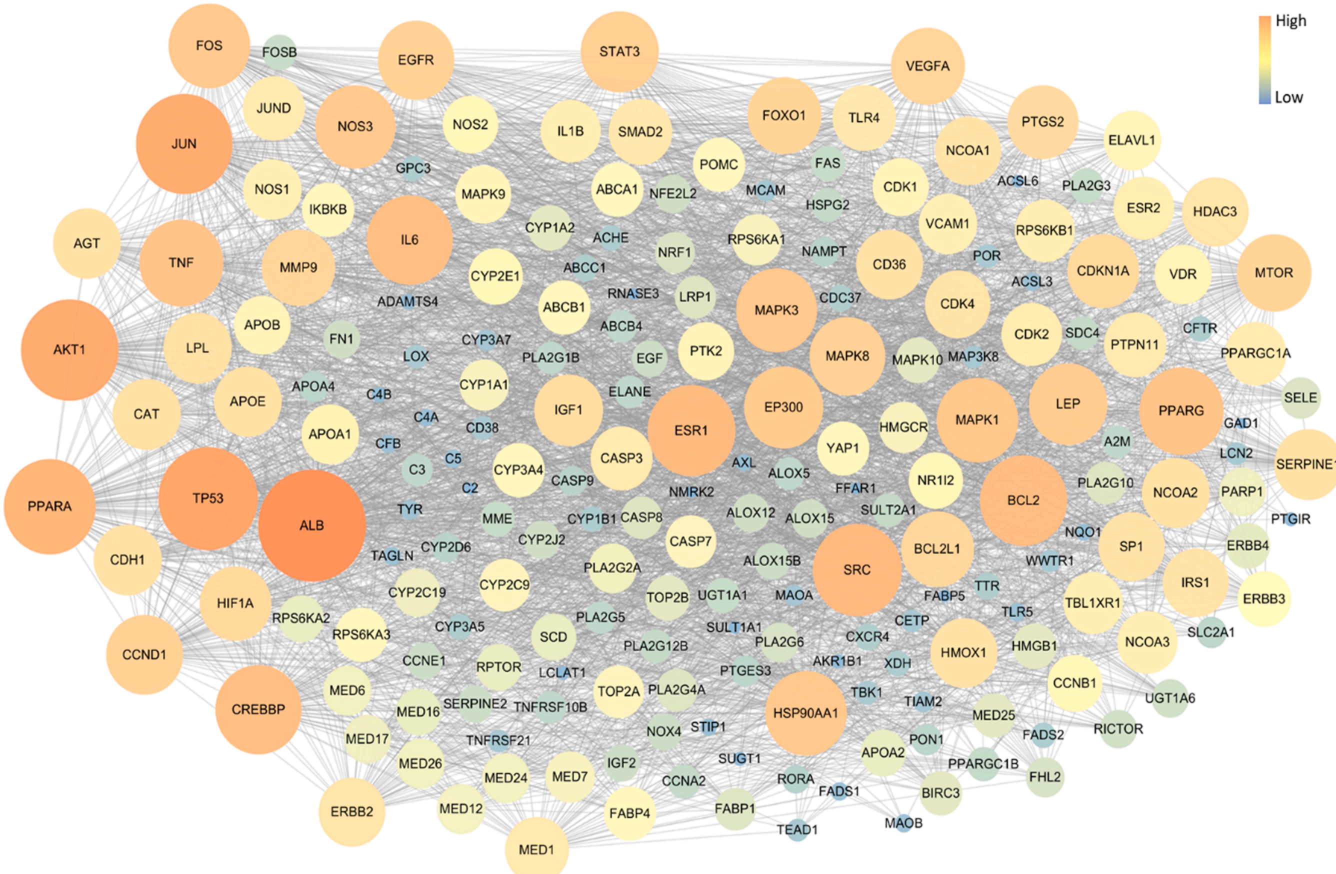

Fig. 8. Topology analysis of PPI: the gene with a large circle and orange color were linked to many genes.

3.4. Potential target genes

Total 373 genes linked to active compounds were searched for using STITCH database, of which only seven active compounds (linoleic acid, luteolin, hederagenin, apigenin, oleanolic acid, isobutyl ohthalate, and pytol) were associated with 373 target genes (Supplementary Table S3). A total of 7531 asthma-related gene were searched in the GeneCards database (Supplementary Table S4). There were total of 202 genes that intersected the 7531 genes involved with asthma and the 373 target genes related with the active compounds. Linoleic acid, apigenin, luteolin were important nodes because these compounds were related with many genes. CASP3, CYP1A1, CYP1A2, CYP1B1, CYP3A4, and TNF were important genes because these genes were associated with three more active compounds (Fig. 7).

3.5. Protein-Protein interaction

The PPI network of 202 potential asthma-associated genes was constructed as 209 nodes and 3467 edges (Fig. 8). As results of network topology analysis, ALB, TP53, AKT1, JUN, PPARA, SRC, ESR1, BCL2, IL6, and CREBBP were main node which were the top 10 genes with many nodes.

3.6. Signal pathway analysis

To investigate the mechanisms of asthma effect of *P. umbrosa*, we performed functional annotations. Seven pathways were selected as asthma-related pathways that is tumor necrosis factor, phosphatidylinositol 3-kinase, mitogen-activated protein kinase (MAPK), nuclear factor kappa, transforming growth factor beta, Wnt, and adenosine 3′,5′-cyclic monophosphate signaling pathways (Fig. 9).

4. Discussion

As the global incidence of asthma gradually increases, concerns about human health are also increasing. Keeping pace with that, the demand for developing a new drug with lower side effects and higher efficacy for treating asthma is also on the rise [21]. In present study, we explored the antiasthmatic effects of PUE using a murine model of OVA-induced asthma. The PUE exhibited suppression of AHR, eosinophilia, and production of cytokines and IgE, which was accompanied with a decrease in pulmonary inflammation and mucus secretion in animals with asthma. In addition, administration of PUE inhibited phosphorylation of ERK and p65 in asthmatic animals, along with the decrease in MMP-9 expression on lung tissue.

Eosinophilia is considered as a crucial therapeutic index for treating asthma [21]. During the onset of asthma, exposure to allergens causes the production of inflammatory cytokines, such as IL-4, -5, and -13, which result in accumulation and activation of eosinophils [15]. Activated eosinophils release chemokines, cytokines, growth factors, and reactive oxygen species via degranulation, which finally exacerbates asthmatic responses including pulmonary inflammation, mucus production, and AHR [6]. Therefore, inhibiting eosinophilia by suppression of inflammatory cytokines is a critical step in controlling asthma. Our study shows that administration of PUE effectively suppressed eosinophilia in OVA-induced asthmatic animals, along with a reduction in the levels of inflammatory cytokines, which led to a decline in AHR, pulmonary inflammation, and mucus secretion. These findings indicate that PUE effectively suppressed allergic responses in OVA-induced asthmatic animals.

MAPKs are cytoplasmic serine and threonine protein kinases that are associated with signal transduction pathways responsible for regulating various cellular processes [17]. In particular, ERK is involved in cell

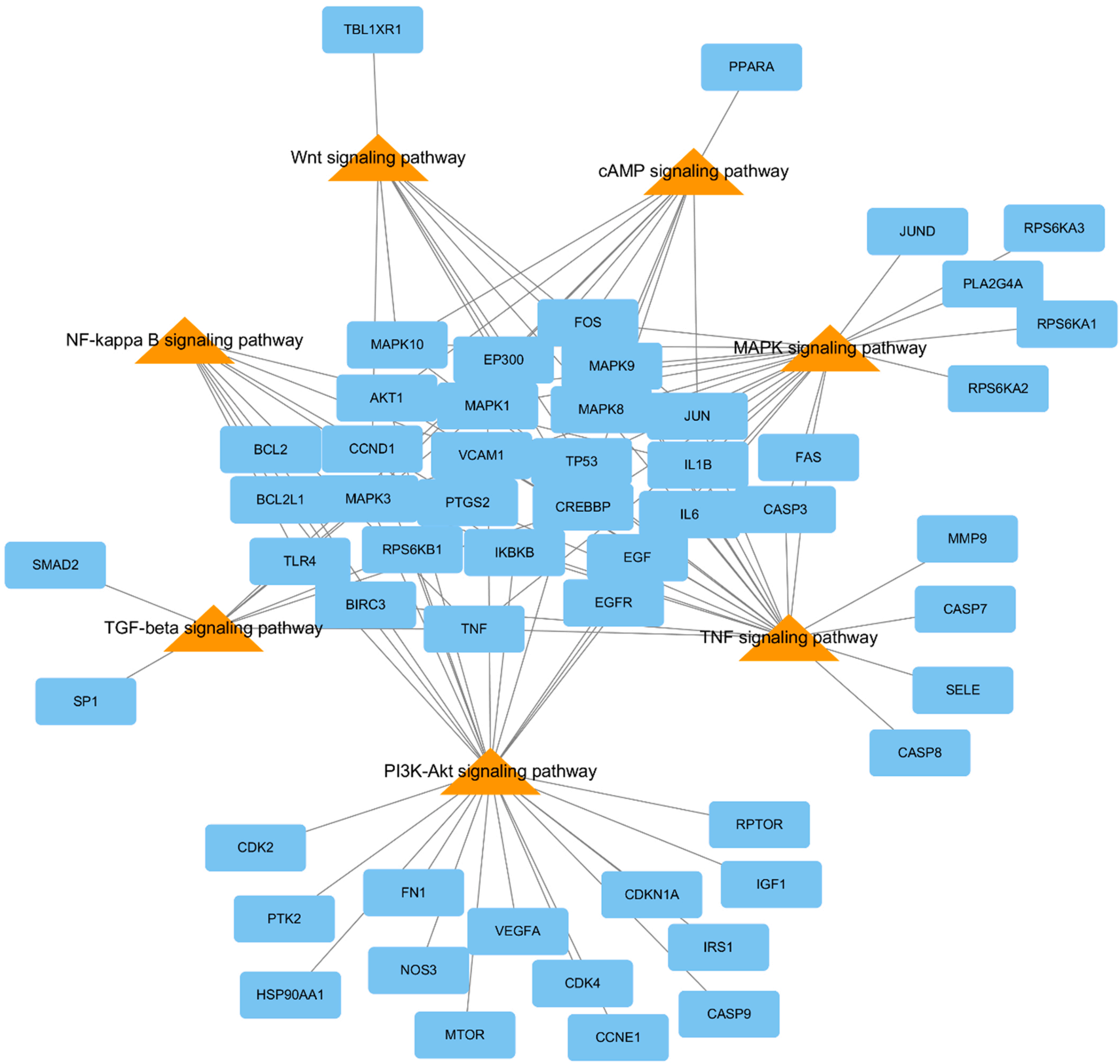

Fig. 9. Target gene-KEGG pathway network of PUE. Genes are shown as cyan rectangles, and pathway are represented by orange triangles.

proliferation, differentiation, and apoptosis through phosphorylation of cytoplasmic components and kinases, and activation of various transcription factors [16]. In the pathophysiology of asthma, ERK is closely associated with allergic responses. Phosphorylation of ERK induces the initial commitment of naive T-helper (Th) cells toward Th2 phenotypes. ERK phosphorylation is also crucial for the survival, maturation, and degranulation of eosinophils, as well as the activation of airway smooth muscle cells, bronchial epithelial cells, fibroblasts, lymphocytes, and mast cells [4]. Thus, these pathways are responsible for increasing cytokines, chemokines, and MMPs, resulting in exacerbation of airway inflammation, AHR, and mucus secretion as seen in the development of asthma [22]. In addition, phosphorylation of ERK induces phosphorylation and nuclear translocation of NF-κB, which acts as a transcriptional activator for several inflammatory mediators such as cytokines, chemokines, and MMPs that promote airway inflammation [27,30]. In contrast, the inhibition of ERK phosphorylation suppressed asthma-induced elevation of eosinophil count and levels of cytokines, chemokines, and IgE, along with the suppression of AHR, airway inflammation, and mucus production, Therefore, ERK is considered as a crucial target molecule in the treatment of asthma [4]. Our study shows that PUE suppressed ERK phosphorylation induced by asthma, inhibited p65 phosphorylation, and suppressed MMP-9 expression, which resulted in the suppression of allergic responses such as eosinophilia, production of cytokines and IgE, AHR, pulmonary inflammation, and mucus secretion. These findings suggested that antiasthmatic effects of PUE is related with the inhibition of ERK phosphorylation.

Herbal medicines exhibit their poly-pharmacological effects by acting on multiple targets due to their multi-component framework [1, 14]. Network pharmacology method has been applied in an attempt to acquire a systemic and more complete understanding of the efficacy and target pathways of herbal medicine [7]. In network pharmacology analysis, *P. umbrosa* was closely associated with the regulation of MAPKs signaling as asthma-related pathway. This evidence supported our findings that antiasthmatic effects of *P. umbrosa* is associated with the suppression of Erk.

In conclusion, PUE effectively inhibited allergic responses in asthmatic animals, which was closely associated with the inhibition of ERK phosphorylation. These findings suggest that *P. umbrosa* may be a potential therapeutic for treatment and management of asthma.

CRediT authorship contribution statement

S.W. Pak contributed to the Methodology, Investigation, Formal analysis, Data curation and Writing – original draft. **A.Y. Lee** contributed to the Methodology, Resources, Formal analysis and Writing – original draft. **Y.S. Seo** contributed to the Formal Analysis and Resources. **S.J. Lee, W.I. Kim, J.S. Kim** and **D.H. Shin** contributed to the Formal analysis. **J.C. Kim** contributed to the Writing – original draft, Writing – review & editing. **J.O. Lim** contributed to the Conceptualization, Formal Analysis, Data curation, Writing – original draft and Writing – review & editing. **I.S. Shin** contributed to the Conceptualization, Supervision, Funding acquisition, Writing – original draft and Writing – review & editing. All authors have read and agreed to the published version of the manuscript.

Conflict of Interest

No competing financial interests exist.

Acknowledgments

This study was supported by the National Research Foundation of Korea grant funded by the Korea Government (Nos. NRF-2019R1A2C4070145 and NRF-2020R1A4A1019395).

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.biopha.2021.112410.

References

[1] N. Chondhary, V. Singh, Insights about multi-targeting and synergistic neuromodulators in Ayurvedic herbs against epilepsy: integrated computational studies on drug-target and protein-protein interaction networks, Sci. Rep. 9 (2019) 10565.
[2] L. Jiang, J. Lu, Y. Qin, W. Jiang, Y. Wang, Antitumor effect of guava leaves on lung cancer: a network pharmacology study, Arab. J. Chem. 13 (2020) 7773–7797.
[3] T.Y. Jung, A.Y. Lee, J.H. Song, M.Y. Lee, J.O. Lim, S.J. Lee, J.W. Ko, N.R. Shin, J. C. Kim, I.S. Shin, J.S. Kim, Scrophularia koraiensis Nakai attnuates allergic airway inflammation via suppression of NF-kB and enhancement of Nrf/HO-1 signaling, Antioxidant 9 (2020) 99.
[4] M. Khorasanizadeh, M. Eskian, E.W. Gelfand, N. Rezaei, Mitogen-activated protein kinases as therapeutic targets for asthma, Pharmacol. Ther. 174 (2017) 112–126.
[5] J.W. Ko, N.R. Shin, S.H. Park, Y.K. Cho, J.C. Kim, C.S. Seo, I.S. Shin, Genipin inhibits allergic responses in ovalbumin-induced asthmatic mice, Int. Immunopharmacol. 53 (2017) 49–55.
[6] O.K. Kwon, J.W. Lee, X. Xuezhen, D.S. Harmalkar, J.G. Song, J.W. Park, D. Hwang, J.H. Min, J.H. Kim, H.K. Han, H.G. Jeong, K.S. Ahn, K. Lee, DK-1108 exerts anti-inflammatory activity against phorbol 12-myristate 13-acetate-induced inflammation and protect effect against OVA-induced allergic asthma, Biomed. Pharmacother. 132 (2020), 110950.
[7] A.Y. Lee, J.Y. Lee, J.M. Chun, Exploring the mechanism of Gyejibokryeong-hwan against atherosclerosis using network pharmacology and molecular docking, Plant 9 (2020) 1750.
[8] A.Y. Lee, W. Park, T.W. Kang, M.H. Cha, J.M. Chun, Network pharmacology-based prediction of active compounds and molecular targets in Yijin-Tang acting on hyperlipidaemia and atherosclerosis, J. Ethnopharmacol. 221 (2018) 151–159.
[9] H.S. Lee, I.H. Lee, K. Kang, S.I. Park, T.W. Kwon, S.J. Moon, C.H. Lee, D.Y. Lee, Systems pharmacology study of the anticervical cancer mechanism of FDY003, Nat. Prod. Commun. 26 (2020) 1–15.
[10] H.S. Lee, I.H. Lee, K. Kang, S.I. Park, S.J. Moon, C.H. Lee, D.Y. Lee, A network pharmacology study on the molecular mechanisms of FDY003 for breast cancer treatment, Evid. Based Complement. Altern. Med. 2021 (2021), 3919143.
[11] S.J. Lee, A.Y. Lee, J.O. Lim, J.H. Lee, T.Y. Jung, S.W. Pak, W.I. Kim, Y.S. Seo, J. C. Kim, J.W. Ko, I.S. Shin, Effect of Yijin-tang, an oriental traditional formula, on allergic responses using an ovalbumin-induced murine asthma model, Evid. Based Complement. Altern. Med. 2021 (2021), 5585692.
[12] S. Li, B. Zhang, Traditional Chinese medicine network pharmacology: theory, methodology and application. Chin, J. Nat. Med. 11 (2013) 110–120.
[13] Y.J. Lee, H.I. Choi, Y.C. Kim, H.S. Shin, H.K. You, Effects of dichloromethane fraction of Phlomidis radix on bone formation in human fetal osteoblasts, J. Periodontol. Implant Sci. 33 (2003) 259–269.
[14] H.H. Nam, J.S. Kim, J. Lee, Y.H. Seo, H.S. Kim, S.M. Ryu, B.C. Moon, A.Y. Lee, Pharmacological effects of Agastache rugosa against gastritis using a network pharmacology approach, Biomolecule 10 (2020) 1298.
[15] K. Oeser, J. Maxeiner, C. Symowski, M. Stassen, D. Voehringer, T cells are the critical source of IL-4/IL-13 in a mouse model of allergic asthma, Allergy 70 (2015) 1440–1449.
[16] J.W. Park, I.C. Lee, N.R. Shin, C.M. Jeon, O.K. Kwon, J.W. Ko, J.C. Kim, S.R. Oh, I. S. Shin, K.S. Ahn, Copper oxide nanoparticles aggravate airway inflammation and mucus production in asthmatic mice via MAPK signaling, Nanotoxicology 10 (2016) 445–452.
[17] G. Pearson, F. Robinson, T. Beers Gibson, B.E. Xu, M. Karandikar, K. Berman, M. H. Cobb, Mitogen-activated protein (MAP) kinase pathways: regulation and physiological functions, Endocr. Rev. 22 (2001) 153–183.
[18] X. Shang, J. Wang, M. Li, X. Miao, H. Pan, Y. Yang, Y. Wang, Antinociceptive and anti-inflammatory activities of Phlomis umbrasa Turcz, Fitoterapia 82 (2011) 716–721.
[19] I.S. Shin, K.S. Ahn, N.R. Shin, C.M. Jeon, O.K. Kwon, Y.W. Chin, K. Lee, S.R. Oh, Homoegonol attenuates the asthmatic responses induced by ovalbumin challenge, Arch. Pharm. Res. 37 (2014) 1201–1210.
[20] N.R. Shin, A.Y. Lee, J.H. Song, S. Yang, I. Park, J.O. Lim, T.Y. Jung, J.W. Ko, J. C. Kim, K.S. Lim, M.Y. Lee, I.S. Shin, J.S. Kim, Scrophularia buergeriana attenuates allergic inflammation by reducing NF-kB activation, Phytomedicine 67 (2020), 153159.
[21] N.R. Shin, S.H. Park, J.W. Ko, Y.K. Cho, I.C. Lee, J.C. Kim, I.S. Shin, J.S. Kim, Lobeglitazone attenuates airway inflammation and mucus hypersecretion in a murine model of ovalbumin-induced asthma, Front. Pharmacol. 9 (2018) 906.
[22] N.R. Shin, H.W. Ryu, J.W. Ko, S.H. Park, H.J. Yuk, H.J. Kim, J.C. Kim, S.H. Jeong, I. S. Shin, Artemisia argyi attenuates airway inflammation in ovalbumin-induced asthmatic animals, J. Ethnopharmacol. 209 (2017) 108–115.
[23] T.Y. Shin, S.H. Kim, D.K. Kim, K.H. Lee, J.S. Park, Phlomis umbrosa root inhibits mast cell dependent allergic reactions and inflammatory cytokines secretion, Phytother. Res. 22 (2008) 153–158.
[24] T.Y. Shin, J.K. Lee, Effect of Phlomis umbrosa root on mast cell dependent immediate-type allergic reactions by anal therapy, Immunopharm. Immuntoxicol. 25 (2003) 73–85.
[25] H. Tang, S. He, X. Zhang, S. Luo, B. Zhang, X. Buan, Z. Zhang, W. Wang, Y. Wang, Y. Sun, A network pharmacology approach to uncover the pharmacological mechanism of XuanHuSuo powder in osteoarthritis, Evid. Based Complement. Altern. Med. 2016 (2016), 3246946.
[26] R.W. Wong, A.B. Rabie, E.U. Hagg, The effect of crude extract from Radix Dipsaci on bone in mice, Phytother. Res. 21 (2007) 596–598.
[27] F. Yuan, R. Liu, M. Hu, X. Rong, L. Bai, L. Xu, Y. Mao, H. Hasimu, Y. Sun, J. He, JAX2, an ethanol extract of Hyssopus cuspidatus Boriss, can prevent bronchial asthma by inhibiting MAPK/NF-kappaB inflammatory signaling, Phytomedicine 57 (2019) 305–314.
[28] X.Y. Zhang, J.L. Simpson, H. Powell, I.A. Yang, J.W. Upham, P.N. Reynolds, S. Hodge, A.L. James, C. Jenkins, M.J. Peters, J.T. Lin, P.G. Gibson, Full blood count parameters for the detection of asthma inflammatory phenotypes, Clin. Exp. Allergy 44 (2014) 1137–1145.
[29] Y. Zhang, R. Chen, W. Xu, A network pharmacology-based analysis of Curcuma longa L, Int. J. Clin. Exp. Med. 14 (2021) 164–173.
[30] Q.L. Zhou, T.Y. Wang, M. Li, Y.X. Shang, Alleviating airway inflammation by inhibiting ERK-NF-kappB signaling pathway by blocking Kv1.3 channels, Int. Immunopharmacol. 63 (2018) 110–118.